

INTELLIGENT DISASTER MANAGEMENT & RELOCATION PLATFORM

TECHNICAL SPECIFICATION: PROGRAMMING LANGUAGES, HUMAN LANGUAGES & AI/ML MODELS

Document Purpose: Comprehensive Technical Reference of Languages, ML Classifiers, GIS Routing Engines & Optimization Solvers
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1. Programming & Technical Languages Used and Their Functions

The application architecture leverages a modern multi-tier tech stack combining strongly-typed client-side Web frameworks, high-performance asynchronous Python backend APIs, spatial relational database engines, and Machine Learning model runtimes:

Language / Tech	Category & Layer	Exact Role & What It Does in the Project
TypeScript (TS)	Frontend Development (Typed JS)	Provides static type safety for React 18 UI components, Leaflet GIS map states, API payload schemas, state management, and user authentication tokens.
JavaScript (ES6+)	Client-side Web Logic	Executes dynamic UI rendering, async HTTP client requests (`apiClient.ts`), Chart.js graph visualization, and browser PWA localStorage caching.
Python (3.14 / 3.10+)	Backend & Machine Learning	Powers the core FastAPI server runtime, ML hazard risk model training/inference (`scikit-learn`), spatial geoprocessing (`geopandas`, `shapely`), and linear optimization solvers.
HTML5 & CSS3	Structure & Styling	Defines semantic web document layouts, accessible web forms, Leaflet map viewports, and custom CWC Royal Blue responsive styling rules (`index.css`, TailwindCSS).
SQL (Spatial SQLite / PostGIS)	Database & Spatial Engine	Manages relational schema storage (`sih_disaster.db`), spatial index execution (`GeoAlchemy2`), shelter bed inventories, citizen records, and 24h hydrograph telemetry.

2. Human Spoken Languages Supported (6 Languages) & Localization

To ensure life-saving emergency information is immediately accessible to citizens, local residents, and rescue teams across Southern India, the **Citizen Safety Portal** features 1-click multi-language translation across **6 spoken languages**:

Language Name	Script & Code	Target Audience & Regional Scope	What It Does in the System
English	Latin (EN)	National Command Officials, State Disaster Chiefs, Tourists	Default administrative portal language for command control, API logs, system health diagnostics, and legal advisories.
Malayalam (Malayalam)	Malayalam (ML)	Wayanad & Kerala state citizens, local panchayats	Translates CWC flood warnings, landslide evacuation notices, and emergency shelter bed availability into Malayalam.
Hindi (Hindi)	Devanagari (HI)	NDRF National Rescue Operations, Central Officials	Provides Hindi translation for national rescue teams, central disaster relief forces, and interstate field ops captains.
Tamil (Tamil)	Tamil (TA)	Bordering Nilgiris / Tamil Nadu district residents	Translates hazard bulletins and helpline emergency numbers for Tamil-speaking populations in adjacent hill sectors.
Kannada (Kannada)	Kannada (KN)	Bordering Mysuru / Kodagu / Karnataka residents	Delivers disaster advisories and safe detour navigation routes to Kannada-speaking evacuees in border sectors.

Telugu (Telugu)	Telugu (TE)	Migrant workforce, regional rescue teams & evacuees	Full portal localization for emergency bulletins, SOS triggers, shelter addresses, and toll-free helpline numbers.
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3. Machine Learning & AI/Mathematical Models Used and What They Do

The platform incorporates **4 core Artificial Intelligence, Machine Learning, and Mathematical Optimization models** to transform raw telemetry and spatial data into actionable emergency evacuation decisions:

AI / ML / Math Model	Algorithm & Technology	Inputs & Processing Logic	What It Does & Output Generated
1. Hazard Risk Classifier Model	Random Forest & Gradient Boosted Decision Trees (`scikit-learn`)	Inputs: 24-hour rainfall (mm), rainfall intensity (mm/h), terrain slope angle (°), soil erodibility K-factor, land-use cover type.	Calculates a threat probability score (0.0 to 1.0) and classifies risk into 4 threat levels (`LOW`, `MODERATE`, `HIGH`, `CRITICAL`). Generates red zone hazard polygons.
2. Spatial Evacuation Routing Engine	Dijkstra Graph Shortest Path & Haversine Geodesic Model (`NetworkX`, `Shapely`)	Inputs: User current GPS lat/lon, shelter lat/lon, road network nodes, active hazard red zone polygon boundaries.	Computes the shortest safe evacuation detour path that dynamically bypasses flooded roads and landslide zones. Outputs step-by-step navigation polyline.
3. Relocation Capacity Optimization Solver	Multi-Objective Linear Integer Programming Solver (`scipy.optimize` / Linear Solver)	Inputs: Hazard-affected population groups, vulnerable counts (children/elderly), distance matrix to shelters, shelter bed limits.	Optimally assigns citizen groups to nearest safe relief camps without exceeding capacities. Generates transparent decision rationale explanations for each route.
4. Resource Carrying Capacity Evaluator	Multi-Criteria Weighted Decision Matrix Model (`numpy`, `pandas`)	Inputs: Shelter bed capacity, current occupancy, water supply days, food supply days, medical kit count, power backup, sanitation units.	Evaluates resource sustainability ratings (0% to 100%) and determines whether a shelter can safely accommodate incoming evacuee cohorts without resource exhaustion.

4. Summary Technical Architecture Matrix

System Function / Feature	Primary Language	AI / ML Model Used	Disaster Mitigation Impact
Landslide & Flood Threat Prediction	Python 3.14	Random Forest Risk Classifier	Early warning generation 24 hours prior to disaster occurrence.
Hazard Avoidance Evacuation Routing	Python & TypeScript	Dijkstra Spatial Routing Graph	Guides citizens safely around submerged roads and landslide debris.
Population Relocation Solver	Python	Linear Integer Programming Solver	Prevents shelter overcrowding and prioritizes high-risk children/elderly.
Multi-Language Emergency Bulletins	TypeScript (6 Languages)	Localization Engine (`translations.ts`)	Eliminates language barriers during emergency evacuations.
Carrying Capacity Assessment	Python & SQL	Multi-Criteria Weighted Evaluator	Ensures adequate food, clean water, medical supplies, and sanitation.